DTC P250A:00 [SKYACTIV-D 1.5]

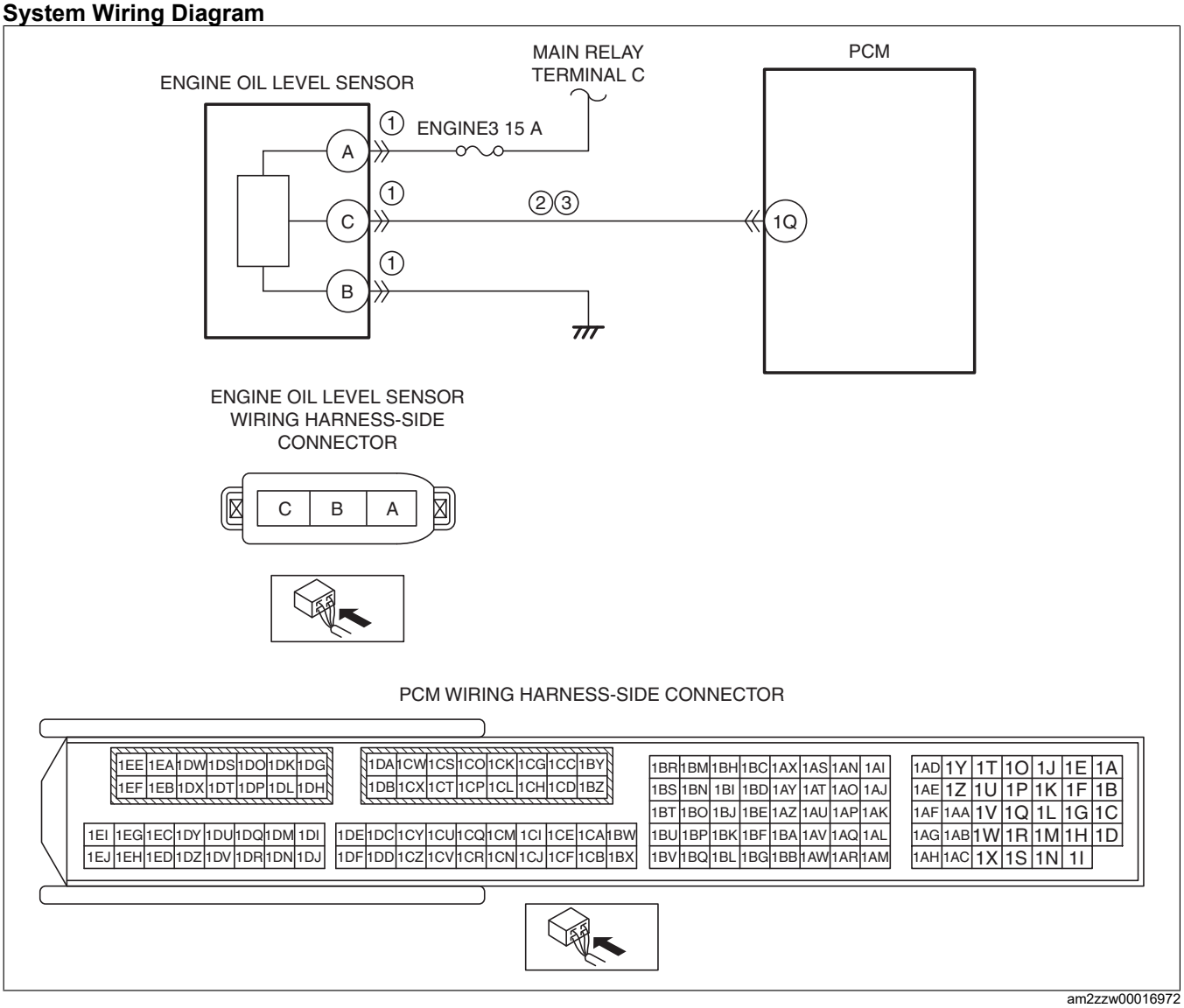
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Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

Details On DTCs

DESCRIPTION	**Engine oil level signal: engine oil level sensor malfunction**	
DETECTION CONDITION	Determination conditions	• PCM receives error signal from engine oil level sensor.
	Preconditions	• Battery voltage: **10—16 V** • The following DTC is not detected: — Communication error between engine oil level sensor and PCM: U1100:00 — Engine oil temperature sensor: P0196:00, P0197:00, P0198:00 • Low-G sensor signal is normal
	Drive cycle	• **1**
	Self test type	• CMDTC self test
	Sensor used	• Engine oil level sensor • Engine oil temperature sensor • Low-G sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Master warning light is illuminated	
POSSIBLE CAUSE	• Excessive engine oil level (oil level exceeds measurable upper limit range of engine oil level sensor) — Poor drainage during engine oil replacement — Excessive engine oil resupplying — Poor oil level adjustment — Engine oil dilution **Note** • If a condition in which the engine oil temperature does not increase sufficiently is repeated, such as when driving short distances, the fuel does not vaporize, engine oil dilution continues, and then the oil level increases. • Engine oil level sensor malfunction — Clogging of oil passage to engine oil level sensor detection area — Engine oil level sensor internal application specific integrated circuit malfunction • Erratic signal to PCM — Engine oil level sensor connector or terminals malfunction — Open or short circuit in wiring harness between engine oil level sensor terminal C and PCM terminal 1Q • PCM malfunction	

System Wiring Diagram



am2zzw00016972

Function Explanation (DTC Detection Outline)

- The PCM receives the signal from the engine oil level sensor and diagnoses the following malfunctions.
  - — With an engine oil temperature of **−35 °C {−31 °F} or more**, the PCM receives an error signal related to the engine oil temperature from the engine oil level sensor continuously for the specified time or more.
  - — The PCM receives an error signal from the engine oil level sensor continuously for the specified time or more.
  - — The oil level signal of the engine oil level sensor received by the PCM does not change for the specified time or more under the following certain diagnostic conditions.
- The PCM performs diagnosis when each of the preconditions is met during a drive cycle. If any of the above malfunctions is detected, a malfunction is determined, DTCs are stored and the master warning light is turned on.

Repeatability Verification Procedure

1. Start the engine and run it at idle.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
EOL	Engine oil level	mm, in	• Displays the engine oil level.
EOT	Engine oil temperature	°C, °F	• Displays engine oil temperature

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P250A:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY OTHER RELATED DTCs OCCURRING** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the other PENDING CODE/DTC also present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**PURPOSE: DETERMINE IF CAUSE IS ERROR SIGNAL FROM ENGINE OIL LEVEL SENSOR** • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Access the following PID using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EOT • Start the engine and warm up until the value of PID EOT is **60 °C {140 °F} or more**. • After engine warm-up, perform the DTC inspection for the PCM. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the next step.
6	**PURPOSE: VERIFY ACTUAL ENGINE OIL LEVEL** • Switch the ignition ON (engine off) and leave for **5 min**. • Pull out the dipstick. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) • Verify if the engine oil level is **8 mm {0.3 in} or more** from the upper limit position of the dipstick. • Is it **8 mm {0.3 in} or more**?	Yes	Go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
7	**PURPOSE: VERIFY IF CAUSE IS ENGINE OIL LEVEL SENSOR STICKING MALFUNCTION** • Access the following PID using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EOL • Start the engine while verifying the monitor value with the ignition switched ON (engine off). • Leave for **30 s or more**. • Does the monitor value decrease by **7 mm {0.3 in} or more** compared to the value with the ignition switched ON (engine off)?	Yes	A temporary malfunction can be considered. • Perform the [ACTION FOR NON-REPEATABLE MALFUNCTIONS]. (See FOREWORD [SKYACTIV-D 1.5].)
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
8	**PURPOSE: VERIFY IF CAUSE IS EXCESSIVE ENGINE OIL LEVEL** • Switch the ignition ON (engine off). • Access the following PID using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EOL • Verify the monitor value with the ignition switched ON (engine off). • Is the monitor value **95.8 mm {3.77 in} or more**?	Yes	Adjust the engine oil level near the middle of the upper and lower limit position on the dipstick. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
9	**PURPOSE: VERIFY IF CAUSE IS ENGINE OIL LEVEL SENSOR STICKING MALFUNCTION** • Access the following PID using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EOL • Start the engine while verifying the monitor value with the ignition switched ON (engine off). • Leave for **30 s or more**. • Does the monitor value decrease by **7 mm {0.3 in} or more** compared to the value with the ignition switched ON (engine off)?	Yes	The cause is an excessive engine oil amount exceeding the measurable range of the engine oil level sensor. Replace the engine oil because there is the possibility that the engine oil is diluted. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Go to the troubleshooting procedure to perform the procedure from Step 4.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1—3
  — Perform the LIN communication line inspection.
- Step 4—5
  — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT ENGINE OIL LEVEL SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the engine oil level sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Go to the next step.
		No	Repair or replace the connector and/or terminals, then go to Step 4.

STEP	INSPECTION		ACTION
2	**PURPOSE: INSPECT ENGINE OIL LEVEL SENSOR CONTROL CIRCUIT FOR SHORT TO GROUND** • Disconnect the PCM connector. • Verify that the engine oil level sensor and PCM connectors are disconnected. • Inspect for continuity between engine oil level sensor terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between engine oil level sensor terminal C and PCM terminal 1Q. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 4.
		No	Go to the next step.
3	**INSPECT ENGINE OIL LEVEL SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the engine oil level sensor and PCM connectors are disconnected. • Inspect for continuity between engine oil level sensor terminal C (wiring harness-side) and PCM terminal 1Q (wiring harness-side). • Is there continuity?	Yes	Replace the engine oil level sensor, then go to the next step. (See ENGINE OIL LEVEL SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Refer to the wiring diagram and verify whether or not there is a common connector between engine oil level sensor terminal C and PCM terminal 1Q. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
4	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
5	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.